

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: MAT101**Course Name: LINEAR ALGEBRA AND CALCULUS****(2019 -Scheme)**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks*

Marks

- 1 Find the rank of the matrix $\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & -4 \\ 0 & 4 & 0 \end{bmatrix}$ (3)
- 2 Find the Eigen values of the matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. What are the Eigen values of A^2 , A^{-1} without using its characteristic equation. (3)
- 3 If $Z = \frac{xy}{x^2+y^2}$, find $\frac{\partial Z}{\partial x}$ and $\frac{\partial Z}{\partial y}$. (3)
- 4 Show that the equation $u(x, t) = \sin(x - ct)$, satisfies wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$. (3)
- 5 Evaluate $\int_0^3 \int_0^2 \int_0^1 xyz \, dx \, dy \, dz$. (3)
- 6 Find the mass of the lamina with density $\delta(x, y) = x + 2y$ is bounded by the x -axis, the line $x = 1$ and the curve $y^2 = x$. (3)
- 7 Find the rational number represented by the repeating decimal 5.373737 ... (3)
- 8 Examine the convergence of $\sum_{k=1}^{\infty} \frac{k^2}{2k^2+3}$ (3)
- 9 Find the Taylor series expansion of $f(x) = \sin \pi x$ about $x = \frac{1}{2}$ (3)
- 10 If $f(x)$ is a periodic function with period 2π defined in $[-\pi, \pi]$. Write the Euler's formulas a_0, a_n, b_n for $f(x)$. (3)

PART B

Answer one full question from each module, each question carries 14 marks.

MODULE 1

- 11 a Solve the following linear system of equations using Gauss elimination method (7)

$$\begin{aligned}x + 2y - z &= 3 \\ 3x - y + 2z &= 1 \\ 2x - 2y + 3z &= 2\end{aligned}$$

- b Find the eigenvalues and eigenvectors of $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ (7)

- 12 a Solve the following linear system of equations using Gauss elimination method. (7)

$$\begin{aligned}2x - 2y + 4z &= 0 \\ -3x + 3y - 6z + 5w &= 15 \\ x - y + 2z &= 0\end{aligned}$$

- b Find the matrix of transformation that diagonalize the matrix $A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$. (7)

Also write the diagonal matrix.

MODULE 2

- 13 a The length and width of a rectangle are measured with errors of at most 3% and 4% respectively. Use differentials to approximate the maximum percentage error in the calculated area. (7)

- b Find the local linear approximation L of $f(x, y, z) = xyz$ at the point $P(1, 2, 3)$. (7)
Compute the error in approximation f by L at the point $Q(1.001, 2.002, 3.003)$.

- 14 a If $w = f(P, Q, R)$ where $P = x - y$, $Q = y - z$, $R = z - x$ prove that (7)

$$\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0.$$

- b Locate all relative extrema and saddle points of $f(x, y) = 4xy - x^4 - y^4$ (7)

MODULE 3

- 15 a Find the area bounded by the parabolas $y^2 = 4x$ and $x^2 = \frac{y}{2}$. (7)

- b Evaluate $\int_{-2}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} e^{-(x^2+y^2)} dx dy$ using polar coordinates. (7)

- 16 a Evaluate $\int_0^1 \int_y^1 \frac{x}{x^2+y^2} dx dy$ by reversing the order of integration. (7)

- b Use triple integral to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ and $x + z = 5$. (7)

MODULE 4

- 17 a Test the convergence of (i) $\sum_{k=1}^{\infty} \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2}$ (ii) $\sum_{k=1}^{\infty} \frac{k^k}{k!}$ (7)
- b Test whether the series $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{\sqrt{k+1}}$ is absolutely convergent or conditionally convergent (7)
- 18 a Test the convergence of the series $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \frac{1}{4.5} + \dots$ (7)
- b Test the convergence of (i) $\sum_{k=1}^{\infty} \left(\frac{k}{k+1}\right)^{k^2}$ (ii) $\sum_{k=1}^{\infty} \frac{7^k}{k!}$ (7)

MODULE 5

- 19 a Find the Fourier series expansion of $f(x) = x - x^2$ in the range $(-1, 1)$. (7)
- b Obtain the half range Fourier cosine series of $f(x) = e^{-x}$ in $0 < x < 2$ (7)
- 20 a Find the Fourier series expansion of $f(x) = x^2$ in the interval $-\pi < x < \pi$. (7)
- Hence show that $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$
- b Obtain the half range Fourier sine series of $f(x) = \begin{cases} x, & 0 < x < \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} < x < \pi \end{cases}$ (7)
